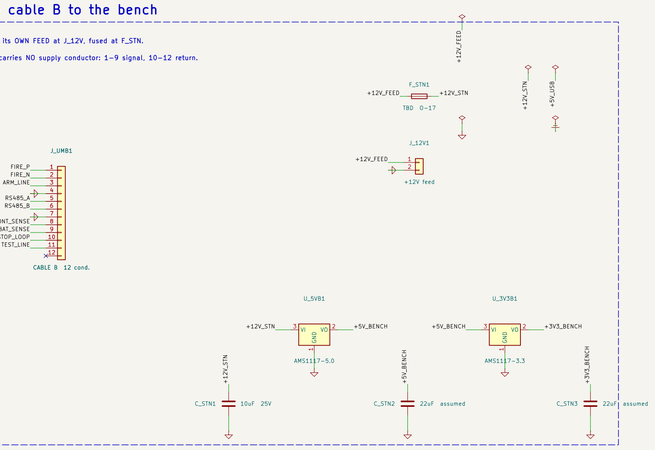
1. UMBILICAL — cable B to the bench

The station supply arrives on its OWN FEED at J_12V, fused at F_STN.

The 12-conductor umbilical carries NO supply conductor: 1–9 signal, 10–12 return.



Bench-side rails: 5 V for the 6N137, 3.3 V for the SP3485.
C_STN1–4 added: this board previously had NO rail decoupling at all.
22 uF values are ASSUMED from the AMS1117 stability note — confirm (0–20).

2. CONTROLS — mechanical, panel-mounted

All four are PANEL controls wired to these terminals, not PCB parts.



S_ARM key: 1–2 ON / 1–3 OFF (pole 1).
4–5 and 6–7 CLOSED IN OFF ONLY.
ARM and TEST can never both be fed.



S_FIRE guarded: 1–2 and 3–4 NO, 5–6 NC aux.
Dead unless ARMED (fed from ARM_LINE).
SKRPABE010 NOT acceptable (0–044).

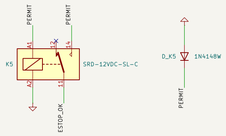
S_ESTOP: 1–2 drops bench K4 (ch 1).
3–4 kills the ARM permit (ch 2).
K1/K2/K3 drop, the shunt re-closes.



S_TEST: momentary ONLY, never latching (H10).
Fed in key-OFF, so dead while ARMED.

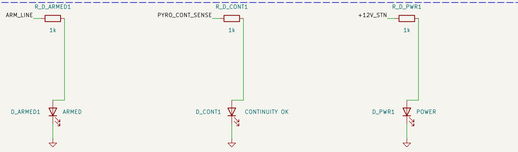


S_RESET sets the permit only with E-STOP released, FIRE released and the key OFF.



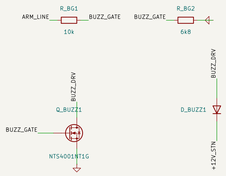
K5 ARM PERMIT: latches via its own NO.
E-STOP release does NOT re-arm.

3. INDICATORS & AUDIO



ARMED lamp driven from the REAL arm line (D-017).

CONTINUITY lamp is ACTIVE-HIGH: the bench sources it, only during TEST and only while DISARMED. LIT = current through the match. A sense-line short to ground reads 'no match' (safe).



Audible ARMED warning — the manual relies on visual cues only.

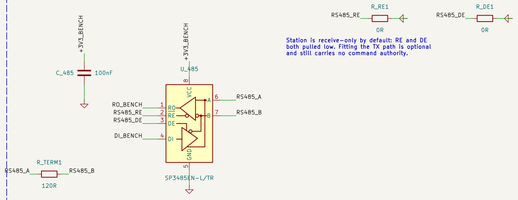
TBD values await D-2 (load cell) and D-6 (isolator part). ARM / FIRE / E-STOP carry no firmware.

4. TELEMETRY — RS-485 in, isolated, out to USB

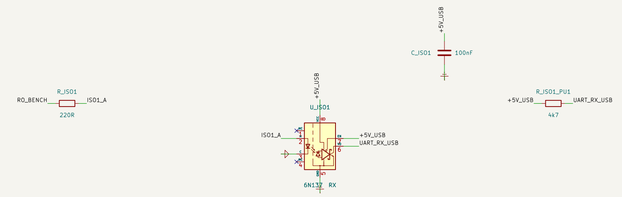
ISOLATION BARRIER down the middle of this block.

LEFT = bench-referenced (GND). RIGHT = laptop side (GNDREF).

No isolated DC-DC: the laptop's USB supplies the isolated side (D-056).



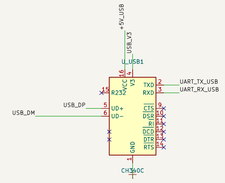
Station is receive-only by default: RE and DE both pulled low. Fitting the TX path is optional and still carries no command authority.



0-21 OPEN, NOT FIXED HERE: this RX path INVERTS.
LED conducts on RO HIGH and an enabled 6N137 pulls its output LOW, so an idle mark reaches the CH340C as a continuous break. Telemetry cannot work as drawn. The fix is a part change, not a rewire — see REV-A1-CHANGES.

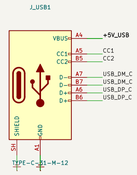
6N137 is fast enough for 115200 baud. The TLP521 / PC817 already in stock are NOT — do not substitute (D-6).

5. USB-C — laptop side, GNDREF domain









CC pull-downs 5k1 mark this as a UFP sink.
Data series 22R, one ESD diode per line.

6. MEASUREMENT OUTPUT — works with no software at all

PATH 2 IS DEFERRED FOR REV A. The SD log is the measurement.



J_DAQ IS NOT FITTED IN REV A (D-19).
It was wired to umbilical pins 8/9, which reach NOTHING on the bench: no buffer, no source. A connector present at both ends of a dead pair reads as a working analog backup, so it is unpopulated and its pins are flagged rather than left inviting a DAQ.
Pin 3 also exported bench GND to the laptop bench, which bridges the isolation barrier whenever a mains-powered DAQ shares it.
Loss of path-2 redundancy for Rev A is recorded in REQUIREMENTS.md.



Battery voltmeter: the pre-test >= 12 V check (manual p. 12), read without walking to the bench.

Review owner: Emilio Guadarrama. Generated by hardware/tools/gen_station.py
TWO GROUND DOMAINS: GND = bench-referenced, GNDREF = laptop/USB. Never joined.
PRELIMINARY — NOT APPROVED: ARM, FIRE and E-STOP are MECHANICAL panel-mounted.
Ignitia Rocket Lab
Sheet: /
File: talavera-station.kicad_sch
Title: TALAVERA v1 — Control Station Board
Size: A1 Date: 2026-11-10 Rev: A.2
KiCad 9.0.8 1 of 1